Machine Learning & Deep Learning Cheat Sheet: Metrics, Loss Functions, and Optimizers

Task	Loss Function	Optimizer	Metric
Binary Classification	Binary Cross Entropy	Adam / AdamW	F1, ROC-AUC, Precision, Recall
Multi-class Classification	Categorical Cross Entropy	Adam / SGD + Momentum	Accuracy, F1-macro
Regression	MSE / MAE / Huber	RMSProp / Adam	RMSE, MAE, R^2
Clustering	Contrastive / Triplet Loss	Adam / SGD	Silhouette, ARI
NLP Tasks	Cross Entropy / KL Divergence / CTC	AdamW / Adafactor / Nadam	BLEU, ROUGE, Perplexity
Computer Vision	Dice + Focal / Cross Entropy	SGD + Momentum / AdamW	IoU, mAP, Dice
Time Series	MSE / Quantile / Huber	Adam / RMSProp	MAPE, RMSE, MASE

Note: Loss functions guide learning, optimizers control convergence, and metrics evaluate performance. Choose based on task, data imbalance, and interpretability needs.